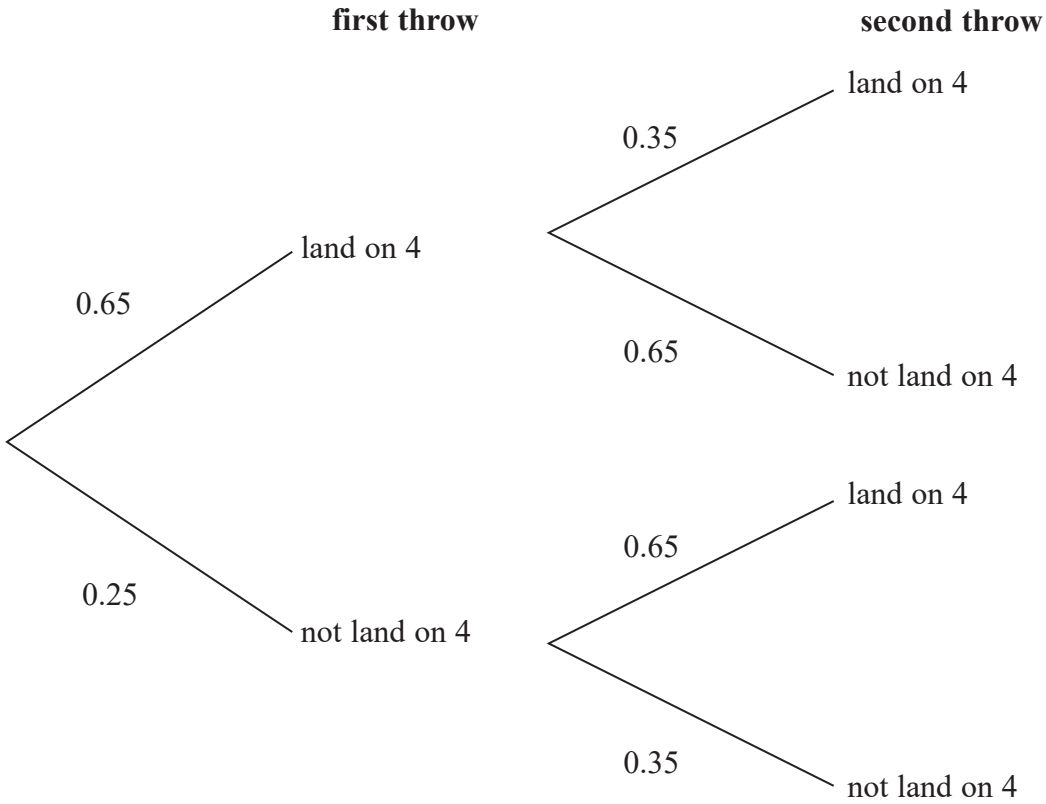
22 When a biased 6-sided dice is thrown once, the probability that it will land on 4 is 0.65
The biased dice is thrown twice.

Amir draws this probability tree diagram.
The diagram is **not** correct.



Write down **two** things that are wrong with the probability tree diagram.

1 *First throw: 0.65 + 0.25 = 0.90, but the probabilities must total 1.*

The 0.25 should be 0.35.

1 mark
Use complementary probabilities

2 *On the upper pair of second-throw branches, 0.35 and 0.65 are swapped.*

P(4) stays 0.65 and P(not 4) stays 0.35 on each throw.

(Total for Question 22 is 2 marks)

1 mark
Recognise the reversed probabilities for the second throw